3.2 Properties of Determinants

Theorem: Row Operations

Let A be a square matrix

a. If a multiple of one row of A is added to another row to produce a matrix B, then $\det B = \det A$.

b. If two rows of A are interchanged to produce B, then $\det B = -\det A$.

c. If one row of A is multiplied by k to produce B, then $\det B = k \cdot \det A$.

Exc Compute $\det A$ where $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$

Sol. Try to reduce A to echelon form:

$$\det A = \begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{vmatrix} \overset{2R_1 + R_2}{=} \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ -1 & 7 & 0 \end{vmatrix} \overset{R_1 + R_3}{=} \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{vmatrix}$$

An interchange of rows 2 and 3 reverses the sign of the determinant. So

$$\det A = -\begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix} = -(1)\cdot(3)\cdot(-5) = 15$$

Ex: Compute $\det A$, where $A = \begin{bmatrix} 2 & -8 & 6 & 8 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{bmatrix}$

Sol.

$$\det A = 2 \cdot \begin{vmatrix} 1 & -4 & 3 & 4 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{vmatrix} = 2 \cdot \begin{vmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & -12 & 10 & 10 \\ 0 & 0 & -2 & 2 \end{vmatrix}$$

$$= 2 \cdot \begin{vmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & -3 & 2 \end{vmatrix} = 2 \begin{vmatrix} 1 & -4 & 1 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

$$= 2 \cdot (1)(3)(-6)(1) = -36 //$$

Theorem: A square matrix A is invertible iff $\det A \neq 0$.

Ex: Compute $\det A$ where $A = \begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ -6 & 7 & -7 & 4 \\ -5 & -8 & 0 & 9 \end{bmatrix}$

Sol

$$\det A = \det \begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ 0 & 5 & -3 & -6 \\ -5 & -8 & 0 & 9 \end{bmatrix} \xrightarrow{2R_1 + R_3} = \begin{vmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ 0 & 0 & 0 & 0 \\ -5 & -8 & 0 & 9 \end{vmatrix} = 0.$$

Ex: Compute $\det A$, where $A = \begin{bmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ -2 & -5 & 4 & -2 \end{bmatrix}$

Sol.

$$\det A = \begin{vmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & -3 & 1 \end{vmatrix} = -2 \begin{vmatrix} 1 & 2 & -1 \\ 3 & 6 & 2 \\ 0 & -3 & 1 \end{vmatrix} = -2 \cdot \begin{vmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & -3 & 1 \end{vmatrix}$$

$-3R_1 + R_2$

$$= -2 \cdot (-1) \cdot \begin{vmatrix} 1 & 2 & -1 \\ 0 & -3 & 1 \\ 0 & 0 & 5 \end{vmatrix} = -2 \cdot -1 \cdot 1 \cdot -3 \cdot 5 = -30 //$$

Theorem: If A is an $n \times n$ matrix, then $\det A^T = \det A$.

Theorem: If A and B are $n \times n$ matrices, then $\det(A \cdot B) = (\det A) \cdot (\det B)$

Ex: Verify above theorem for $A = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$

Sol

$$A \cdot B = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix} \cdot \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 6 \cdot 4 + 1 \cdot 1 & 6 \cdot 3 + 1 \cdot 2 \\ 3 \cdot 4 + 2 \cdot 1 & 3 \cdot 3 + 2 \cdot 2 \end{bmatrix}$$

$$= \begin{bmatrix} 25 & 20 \\ 14 & 13 \end{bmatrix}. \quad \det(A \cdot B) = 25 \cdot 13 - 20 \cdot 14 = 45 //$$

Since $\det A = 6 \cdot 2 - 1 \cdot 3 = 9$ and $\det B = 4 \cdot 2 - 3 \cdot 1 = 5$

$\det A \cdot \det B = 45 // = \det(A \cdot B)$.